

A quantitative refinement of the seven-eighths zero-free half-plane

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Abstract

We refine OpenAI’s zero-free half-plane for all Dirichlet L -functions and all finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$. The boundary is $2187392879/2500000000 = 0.8749571516$, with principal poles allowed. The proof is conditional on OpenAI’s unrefereed preprint: it uses its seven-eighths theorem as an input and reuses its construction and estimates under modified parameters, with a hypothesis audit for each reused statement. We first change the averaging scales in its cubic-theta probe and obtain $20999/24000$. We then extend the range of its plain-moment estimate and let the prime-slot capacity use this improved boundary. Two finite applications give the stated result. The proof retains the residue, tail, height and normalization costs; the recursive moment argument is supplied in a companion proof note.

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1 Introduction

Throughout, $F = \mathbb{Q}(\sqrt{-3})$. OpenAI [2] gives a proof, in an unrefereed preprint, that every Dirichlet L -function and every finite-order Hecke L -function over F is zero-free in $\operatorname{Re} s > 7/8$. Its proof compares a cubic-theta character sum with a Mellin signal containing $1/L$. We retain this construction, change the averaging scales, and verify that its plain moment can use a slightly wider parameter range. The latter allows the comparison to use the boundary already obtained in the preceding step.

Theorem 1.1. *Put $\theta_* = 2187392879/2500000000 = 0.8749571516$. Every finite-order Hecke L -function over F and every Dirichlet L -function, including imprimitive presentations, is zero-free in $\operatorname{Re} s > \theta_*$, with principal poles at $s = 1$ excepted. In particular this holds for the Riemann zeta function. Every zero ρ with $0 < \operatorname{Re} \rho < 1$ of one of these L -functions satisfies*

$$\frac{312607121}{2500000000} \leq \operatorname{Re} \rho \leq \frac{2187392879}{2500000000}.$$

The boundary is independent of the conductor and height. Constants and thresholds in the character-sum estimates may depend on the fixed target. The proof uses [2, Theorem 1.1], namely $\beta_* \leq 7/8$ in the notation of Section 5, as a hypothesis, together with the statements of [2] listed in Table 1. Theorem 1.1 is therefore conditional on the correctness of those statements.

The proof begins with the following intermediate boundary.

Proposition 1.2. *Under the same external inputs, every member of the stated families is zero-free in $\operatorname{Re} s > \theta_0 = 20999/24000$, with principal poles at one excepted.*

For Sections 2–6, write $\theta = \theta_0$. The extra prime length lowers the direct estimate by $1/24000$ while leaving the principal exponent $s - 11/16$ unchanged. It also changes the estimate for the remaining rows. Retaining the perturbation in OpenAI’s two-parameter endpoint certificate [2, Lemma 20.2] gives a quadratic that is positive throughout the parameter rectangle. The resulting margin pays the change of scales and the recovery errors. Section 7 then proves Theorem 1.1 by extending the plain-moment range and using each established all-height boundary as input to the next step. The construction never uses the output boundary in its own moment estimate.

Sections 2–4 establish the low estimate. Section 5 proves the endpoint bound, and Section 6 completes the first-stage recovery and continuation. Section 7 gives the two adaptive steps; the full recursive moment proof is in [1]. Section 8 records the limits of the parameter refinement and the scope of the verification archive. Appendix A audits the hypotheses of every reused statement.

The reused inputs are listed in Table 1. The new arguments are the modified-scale low estimate, the re-run endpoint certificate at the perturbed geometry, the admission of the principal Euler correction below $7/8$, and the extended plain-moment range with its finite feedback. No improved long inverse moment is assumed or proved here.

Table 1: Statements of [2] used in this paper. Each is audited in Appendix A.

Statement in [2]	Use here	Section
Theorem 1.1 ($\beta_* \leq 7/8$), as a hypothesis	Bin ceiling $\delta \leq 3/4$; Lemma 18.1 at $\kappa = 3/4$; small-row line $\operatorname{Re} s = 7/8 + 8e$ right of β_*	5, 6
§§ 4–7 and 16: probe (6.1), separation (6.2), local identity (7.11)–(7.12), Poisson identity and tuple correction (16.3), (16.8), Lemma 16.2	Character sum, high identity, local factors	2, 6
Lemmas 4.5, 4.6, 4.9, 4.10	Annular tails; logarithmic control; deleted Euler factors	3–6
Lemma 14.3; proof of Lemma 15.1	Reflected energy; Lemma 3.1	3
Proposition 15.2; proof of Proposition 15.3	Gram bound; Proposition 4.1	4
Proposition 8.3; Lemmas 17.1, 17.6, 18.1; proof of Proposition 19.2	Witness counts at $\widehat{\kappa} = 3/4$	5
Lemmas 10.3, 10.4, 20.1; (20.4), (20.8); certificate of Lemma 20.2; (20.5), (20.7), (20.11)	Central contours, endpoint exponent, Lemma 5.1, remaining branches	5
Lemma 13.1; Lemma 10.5 with (20.1)–(20.3); Lemma 10.6 with §20.3; Lemma 11.1 with §20.6	Normalizer, principal recovery, outer rows, height allocation	6
Proposition 2.1 (proof repeated); Proposition 11.3	Continuation and transfer to the full family	6
Proof of Lemma 18.1, extended in [1]	Plain capacity at κ_0 and κ_1 ; each uses the previous all-height boundary	7

2 The probe and its averaging scales

2.1 Field, characters and calibration

Let $F = \mathbb{Q}(\sqrt{-3})$, $\omega = e^{2\pi i/3}$ and $\lambda = 1 + 2\omega$. Its ring of integers is $\mathcal{O}_F = \mathbb{Z}[\omega]$. For an element or ideal write $q_a = N(a)$; for a nonzero element put $\alpha(a) = a/|a|$. Good ideals are represented by their unique primary generators, congruent to one modulo 3.

Fix a primitive finite-order target character η . The fixed ray group $\mathcal{T}$ presents the supplementary and reciprocity phases G and $\mathcal{R}$ of [2, Section 4]; it is independent of the target. Choose the finite excluded set S to contain the primes over 6, the full defining-modulus supports of η and $\widehat{\mathcal{T}}$, and every prime below a fixed cutoff P_0 . The defining modulus includes redundant primes at which an induced presentation is extended by zero. This entire datum is fixed before the scale Z and moving prime labels are chosen.

Let $I_\eta(X, Y, Z)$ be exactly the calibrated probe of [2, (6.1), Section 6.1], with nonnegative nonzero annular tests W_0, W_1 and Gaussian Mellin weight e^{ι^2} . We retain its auxiliary factors $b_*, \xi, \tau, \Xi, G, \mathcal{R}$ and normalized cubic Gauss sum γ_2 . Here b_* generates the squarefree product of S ; $\xi \bmod b_*$ is primitive, of order dividing six and nonprincipal at every prime factor, $|\tau| = 1$, and $\Xi(a) = \xi(a)\chi_a(b_*)$. The sextic symbol $\chi_a(v) = (v/a)_6$ has its original zero on nonunits. Thus a sixth power of the symbol is a coprimality indicator. Inverses of Ξ are taken only on units, and no character is replaced by its primitive presentation.

The completed indices are cn^3 , with c squarefree and n unrestricted; they may share primes. All ideal variables avoid S . For a squarefree product D of slot primes, define $I_{\eta;D}$ by inserting $\mathbf{1}_{D|cn^3}$ in this same completed row. The auxiliary phases and the weights $q_c^{-1/2}q_n^{-1}$ remain unchanged. This convention fixes the source in both the direct estimate and its Poisson representation.

2.2 Compensation and geometry

Take an even fixed slot count K , positive lengths ℓ_i with $\sum_i \ell_i = \ell$, and nonnegative nonzero annular weights W_i . Put $P_i = Z^{\ell_i}$. The underlying prime windows are disjoint before ray or excluded-set restrictions. Set

$$\mathcal{P}_i(Z) = \{p \notin S : p \text{ prime}, p \in 1_{\mathcal{T}}, q_p/P_i \in \text{supp } W_i\}.$$

For a tuple $\mathbf{p}$ and subset $J \subseteq \{1, \dots, K\}$, put $p_J = \prod_{i \in J} p_i$. Use the same finite operation as [2, (12.5)]:

$$\mathcal{I}_{\eta, \text{mod}}(Z) = \sum_{\mathbf{p} \in \prod_i \mathcal{P}_i(Z)} \prod_i W_i(q_{p_i}/P_i) \sum_J (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_J^c)} I_{\eta; p_J^c} \left(\frac{X}{q_{p_J}}, \frac{Y}{q_{p_J}}, Z q_{p_J^c} \right). \quad (1)$$

Every window remains at its original P_i . Membership in $1_{\mathcal{T}}$ does not imply $\eta(p) = 1$; the conjugate target factor in (1) is retained.

We choose

$$\begin{aligned} \varpi &= \frac{1}{6000}, & b &= \frac{1}{8}, & \ell &= \frac{1}{6} + \varpi, & h &= \frac{13}{16} + \frac{3\varpi}{2}, \\ l_x &= \frac{17}{48} - \frac{\varpi}{2}, & l_y &= \frac{23}{48} - \frac{\varpi}{2}, & M &= \frac{5}{6} - \varpi, \\ X &= Z^{l_x}, & Y &= Z^{l_y}, & M + \ell &= 1, & l_y - l_x &= b, & h &= 1 - l_x + \ell, \\ \theta &= \frac{7}{8} - \frac{\varpi}{4} = \frac{20999}{24000}. \end{aligned} \quad (2)$$

At $\varpi = 0$ these are the values of [2, (12.4)]; the scalar perturbation ϖ is the only change. The fine slot system is chosen later from the moment mesh. It remains fixed as Z tends to infinity.

For a rescaled subset put $d = \sum_{i \in J} \ell_i$. Write $q_{p_J} = Z^d r_J$, with r_J in one fixed positive compact interval, uniformly over the tuples. Then the actual low scales are

$$X' = Z^{l_x-d}/r_J, \quad Y' = Z^{l_y-d}/r_J, \quad Q = q_{b_*} X' Y' \asymp Z^{M'}, \quad M' = 1 - \ell - 2d, \quad \ell' = \ell - d.$$

In particular $l_x - \ell, l_y - \ell$ and $M - 2\ell$ are positive. All rescaled row and lattice scales eventually exceed one, uniformly in the fixed annular ratios.

2.3 The shared Mellin representation

Let $\widetilde{W}_0(|z|^2)$ be the planar Fourier transform for the self-dual measure $(2/\sqrt{3}) dx dy$, and put

$$M(z) = \int_0^\infty \widetilde{W}_0(r) r^{z-1} dr, \quad \widehat{W}_1(w) = \int_0^\infty W_1(r) r^{w-1} dr.$$

Both transforms have rapid vertical decay in the relevant fixed strips; M is holomorphic for $\operatorname{Re} z > 0$ and $M(1/6) > 0$ by [2, Section 7.1].

The exact Poisson identity of [2, Section 16.1] gives

$$\begin{aligned} \mathcal{I}_{\eta, \text{mod}}(Z) &= \frac{1}{(2\pi i)^3} \int_{(3)} \int_{(3)} \int_{(2)} X^{1/2-z} Z^{s+z-1} Y^{w-1} e^{(s+z-1)^2} M(z) \widehat{W}_1(w) \\ &\quad \times \sum_u^{(6)} q_u^{-z} \overline{\xi(u)} \frac{\zeta_F^S(6z) L_F^S(w, \chi_\bullet(u))}{L_F^S(s, \eta \chi_\bullet(u))} \mathfrak{H}_{\eta, u, Z}(s, w, z) dz dw ds. \end{aligned} \quad (3)$$

The sum includes all nonzero sixth-power-free element rows prime to S , with unit factors. Superscript S denotes deleted Euler factors. The correction $\mathfrak{H}$ is the full quotient-free selected tuple correction of [2, (16.3)]. At a selected prime it uses the holomorphic factor G_p ; all unselected primes retain H_p . This definition is valid also at zeros of individual raw local quotients. Identity (3) concerns exactly the finite physical expression (1).

At the principal double residue $u = 1, w = 1, z = 1/6$, the outside power is $X^{1/3} Z^{s-5/6}$ and the slot factor supplies $Z^{\ell/6}$. Thus the principal exponent of [2, (12.3)] persists:

$$C(s) = s - \frac{5}{6} + \frac{l_x}{3} + \frac{\ell}{6} = \frac{l_x}{2} + s - 1 + \frac{h}{6} = s - \frac{11}{16}. \quad (4)$$

Increasing ℓ therefore leaves $C(s)$ unchanged. The low estimate, by contrast, decreases with l_x ; this is the source of the improvement.

3 The completed row energy

The direct estimate uses the energy of the completed rows after reflection. Lemma 3.1 extends [2, Lemma 15.1], which is stated for $\ell = 1/6$, to $0 \leq \ell \leq 1/5$. Its proof follows the proof of [2, Lemma 15.1], recomputing the exponent from [2, Lemma 14.3]; at $\ell = 1/6$ the bound (6) is [2, (15.2)].

Fix the target datum and slot system before Z . For a rescaled subset J and tuple $(p_i)_{i \in J}$, write

$$\begin{aligned} d &= \sum_{i \in J} \ell_i, & \ell' &= \ell - d, & M' &= 1 - \ell - 2d, \\ q_{p_J} &= Z^d r_J, & X' &= Z^{l_x-d}/r_J, & Y' &= Z^{l_y-d}/r_J, & Q &= q_{b_*} X' Y' \asymp Z^{M'}. \end{aligned} \quad (5)$$

Here r_J lies in a fixed positive compact interval. We use exactly the row $B_{m,\sigma}^J$ defined in [2, §15.1], with ray expansion from [2, §6]. Its surviving prime weights are $\prod_{i \notin J} \eta(p_i) W_i(q_{p_i}/P_i)$, summed inside the row before squaring. Fixed window bounds enter the constant.

These product weights are independent of the row and dual variables. The external squarefree label is one. Additional punctures are fixed, and the moving character $\chi_\bullet(m)$ retains its zeros. Local symbol powers are extended by zero on nonunits also at exponent zero. All nonresidual local primes are frozen; residual rows and active marked products are separated into classes modulo the full fixed reflection modulus. We retain the original zero extensions and dual support, including shared primes. The row ball includes every nonzero element, with its unit and bad-prime sectors.

Lemma 3.1 (cf. [2, Lemma 15.1]). *For a fixed disjoint slot system with $0 \leq \ell \leq 1/5$, and every $\varepsilon > 0$,*

$$\sum_{0 < q_m \ll Q} |B_{m,\sigma}^J(Z)|^2 \ll_{\varepsilon, \text{fixed}} Z^{M' + (5\ell - 1 + d)_+ / 4 + \varepsilon}. \quad (6)$$

The constant and threshold are uniform in the rescaled tuple. They depend on the fixed datum and finitely many window seminorms.

Proof. Use the block variables of [2, §14.4]: O, H are the powerful and residual row lengths; A_0 is the active frozen radical length; N_0, B_0, S_0 describe the forced and small local factors. The active marked length is z_a , and the remaining dual lengths are v, ℓ_b, e_λ . Put

$$T_d = 2H + 2A_0 + 2z_a - N_* - N_0 - 3B_0, \quad u = \min\{v, z_a, (v + z_a)/3\}. \quad (7)$$

The imported bound for a retained block, after summing its frozen parts, is $Z^{E_{\text{ref}} + \varepsilon_0}$, where

$$\begin{aligned} E_{\text{ref}} &= O/2 + \max(H, v + \ell_b) - S_0 - B_0 + z_a - u - \ell_b - 2e_\lambda/3 \\ &\quad - \frac{1}{2}(T_d - v - 3\ell_b - e_\lambda)_+. \end{aligned} \quad (8)$$

It is uniform on bounded length ranges, with finite seminorms and polynomial norm-height cost. The powerful-part count $O/2$ is already included. Unretained whole blocks are arbitrarily power-small, with their finite kernel order chosen before the polynomial row, tuple and local multiplicities are summed.

First take $N_* = 1 + \ell'$ and suppress the small annular errors. The actual low source gives

$$\begin{aligned} \Delta_H &= M' - O - H \geq 0, \quad 2A_0 \leq O, \quad 0 \leq N_0 \leq A_0, \quad S_0, B_0 \geq 0, \\ 0 &\leq z_a \leq \ell', \quad v, \ell_b, e_\lambda \geq 0, \quad v + 3\ell_b + e_\lambda \leq T_d. \end{aligned} \quad (9)$$

Since $M' + \ell' - 1 = -3d$,

$$T_d - (H - 3d) = H - M' + 2A_0 + 2(z_a - \ell') - N_0 - 3B_0 \leq 0. \quad (10)$$

Hence $\max(H, v + \ell_b) = H$.

If $u = z_a$, (8) gives $E_{\text{ref}} \leq O/2 + H \leq M' - \Delta_H$. Otherwise $u \geq v/2$: when the minimum is $(v + z_a)/3$, its comparison with z_a gives $v \leq 2z_a$. Writing $y = v + 3\ell_b + e_\lambda$, we obtain

$$u + \ell_b + 2e_\lambda/3 + \frac{1}{2}(T_d - y)_+ \geq y/4 + \frac{1}{2}(T_d - y)_+ \geq T_d/4.$$

Thus

$$\begin{aligned} E_{\text{ref}} &\leq O/2 + H - S_0 - B_0 + z_a - T_d/4 \\ &\leq M' + (1 + 3\ell' - 2M' - 2\Delta_H)/4 \\ &= M' + (5\ell - 1 + d - 2\Delta_H)/4. \end{aligned} \quad (11)$$

Here $z_a \leq \ell'$ and $2A_0 - N_0 + B_0 + 4S_0 \geq 0$. Both branches therefore give

$$E_{\text{ref}} \leq M' + (5\ell - 1 + d)_+/4. \quad (12)$$

It remains to restore the Gaussian row. For $\varrho_i = q_{p_i}/P_i$ and $r = q_{cn^3}/Z^{1+\ell'}$, the annulus $r = e^k x$ has the joint profile

$$\chi(\log x) W_G \left(e^k x / \prod_{i \notin J} \varrho_i \right) \prod_{i \notin J} W_i(\varrho_i), \quad W_G(y) = \frac{e^{-(\log y)^2/4}}{2\sqrt{\pi}}.$$

Its homogeneous seminorms are summable against every fixed $e^{A|k|}$. Lemmas 4.5–4.6 of [2] discard whole annuli with $|k| > \kappa_G \log Z + C$ at arbitrary power saving and separate each remaining profile through a Fourier density common to all rows and tuples. The prime weights remain product weights, and $q_c^{-1/2} q_n^{-1}$ is unchanged. The Gaussian decay absorbs the polynomial counting factors before these annular tails are discarded.

The completed length is now $1 + \ell' + k/\log Z$. Dyad comparisons and the bounded negative ramified range add $O_{\text{fixed}}(1/\log Z)$ errors. Since the preceding expressions are piecewise affine in finitely many lengths, their total error is $O(\kappa_G + \tau_{\text{ref}} + 1/\log Z)$, where $\tau_{\text{ref}} > 0$ is the retention slack in Lemma 14.3. Choose these allowances inside ε , then enlarge the threshold. The kernel amplitude is extracted at the actual row width before positive sieve enlargement. Summing the logarithmic dyads and finite sectors, followed by Minkowski through the common density and Gaussian annuli, proves the lemma. $\square$

4 Recovery of the physical low estimate

The additive polynomial $A_{m,\sigma,\nu}(Y')$ is the one in the exact separation [2, (6.2)]. Its fixed ray coefficient is $\mathbf{1}_{s \in \sigma} \chi_s(b_*)/(\tau \xi(s))$, extended by zero off the original primary support before evaluating the inverse. For $Q, Y' \geq 1$, in fixed polynomial ranges, and

$$P_a = Y'^2/Q \geq 1, \quad (13)$$

[2, Proposition 15.2] gives

$$\sum_{q_m \ll Q} |A_{m,\sigma,\nu}(Y')|^2 \ll_{\varepsilon, \text{fixed}} (1 + |\nu|)^{J_A} \frac{Q}{Y'} \left(1 + P_a^{1/6} + \frac{P_a^2}{Y'} \right) Z^\varepsilon. \quad (14)$$

The order J_A is independent of the moving scales and height. The same input covers fixed finite combinations of these ray coefficients.

At (5),

$$P_a = q_{b_*}^{-1} Z^b, \quad b = 1/8. \quad (15)$$

For $\varpi = 1/6000$, every $0 \leq d \leq \ell$ satisfies

$$\begin{aligned} M' &\geq \frac{1}{2} - 3\varpi > 0, \\ l_x - d &\geq \frac{3}{16} - \frac{3}{2}\varpi > 0, \\ l_y - d - 11b/6 &\geq \frac{1}{12} - \frac{3}{2}\varpi > 0. \end{aligned} \quad (16)$$

The bounded tuple ratios therefore give $Q, Y', P_a \geq 1$ eventually, uniformly in J , and $Y' \gg P_a^{11/6}$. Both the constant term and P_a^2/Y' in (14) are then $O(P_a^{1/6})$. Hence

$$\sum_{q_m \ll Q} |A_{m,\sigma,\nu}(Y')|^2 \ll (1 + |\nu|)^{J_A} (Q/Y') P_a^{1/6} Z^\varepsilon. \quad (17)$$

The following proposition is [2, Proposition 15.3] at the perturbed geometry; its proof repeats that proof, with Lemma 3.1 in place of [2, Lemma 15.1]. At $\varpi = 0$ the bound is [2, (15.6)].

Proposition 4.1 (cf. [2, Proposition 15.3]). *For the geometry (2) and every $\varepsilon > 0$, the complete compensated probe satisfies*

$$|\mathcal{I}_{\eta, \text{mod}}(Z)| \ll_{\varepsilon, \text{fixed}} Z^{l_x/2+b/12+\varepsilon} = Z^{3/16-1/24000+\varepsilon}. \quad (18)$$

The datum, windows and slot count are fixed before Z ; all subsets and tuples in (1) are included.

Proof. For a fixed rescaled tuple, apply Cauchy in the shared physical rows of [2, (6.2)], whose normalization is $Q^{-1/2}$. Lemma 3.1 and (17) give

$$Q^{-1/2} \left(\frac{Q}{Y'} P_a^{1/6} \right)^{1/2} Z^{M'/2+(5\ell-1+d)_+/8+\varepsilon} \ll (X')^{1/2} P_a^{1/12} Z^{(5\ell-1+d)_+/8+\varepsilon}. \quad (19)$$

We used $Z^{M'}/(X'Y') = r_J^2$, a bounded factor. The polynomial height costs from the two inputs are integrable against the common, rapidly decreasing Mellin density. Choosing its finite derivative order after the tail orders absorbs those costs uniformly in the tuple.

Ideal counting gives $O(Z^{d+\varepsilon})$ tuples on J . Their coefficient is $q_{p_J}^{-3/2} \asymp Z^{-3d/2}$, and $(X')^{1/2} \asymp X^{1/2} Z^{-d/2}$. Their total additional exponent in (19) is

$$d - 3d/2 - d/2 + (5\ell - 1 + d)_+/8 \leq -7d/8 \leq 0, \quad (20)$$

because $\ell < 1/5$. The surviving J^c weights, including $\overline{\eta(p_{J^c})}$, were already summed inside $B_{m,\sigma}^J$.

Sum the finitely many subsets and fixed ray sectors, allocating their small-power losses inside ε . The Gaussian and Mellin integrations have the finite weighted norms used above. Finally, $P_a^{1/12} \asymp Z^{b/12}$ and $l_x/2 + b/12 = 3/16 - \varpi/4$, proving the proposition. $\square$

Since $C(s) = s - 11/16$ in (4), $C(20999/24000) = 3/16 - 1/24000$. Thus the full low estimate matches the boundary of the principal signal.

5 The high estimate and its endpoint

Let β_* be the supremum of $1/2$ and the real parts of nontrivial zeros of primitive finite-order Hecke L -functions over F . By [2, Theorem 1.1], which we use as a hypothesis, $\beta_* \leq 7/8$. Suppose $\beta_* > \theta$, and put $\Gamma = \beta_* - \theta > 0$, $\alpha = 5/6$.

5.1 Balancing the witness counts

For rows $q_u \asymp U = Z^d$, let R be the counting exponent: $\#\mathcal{B} \ll U^{R+\varepsilon}(1+T_1)^A$, with a fixed height order A . Use the detector bins of [2, Section 8], with real parameter a and freely chosen cutoff $t \in [1, 3/2]$. In a nonfloor bin write $\delta = 2a - 1$ and let $q \in [0, \delta/2]$ be the length-weighted prime amplitude; error slots receive amplitude zero. The bin ceiling $a \leq \beta_*$ of [2, Section 8] (compare [2, (12.2)]) and [2, Theorem 1.1] put actual bins in $\delta = 2a - 1 \leq 2\beta_* - 1 \leq 3/4$; we retain the larger closed rectangle $\delta \leq 5/6$ for the endpoint certificate. We repeat the selection proof of [2, Proposition 19.2] with the fixed value $\hat{\kappa} = 3/4$ in place of $\kappa = 2\beta_* - 1$. That proposition is stated under the Part II supposition [2, (12.1)], $\Delta = \beta_* - 7/8 > 0$, and its count carries a loss $+\Delta/4$ from the dynamic plain capacity $(1 - 2m)/(6\kappa)$, $\kappa = 3/4 + 2\Delta$ [2, (19.2)]. Here $\Delta > 0$ is unavailable and unnecessary: the proof uses κ only through the bin ceiling $\delta \leq \kappa$ and [2, Lemma 18.1], whose

zero-free hypothesis at $\hat{\kappa} = 3/4$ is $\beta_* \leq (1 + \hat{\kappa})/2 = 7/8$; both are supplied by [2, Theorem 1.1]. With $\hat{\kappa}$ fixed the plain capacity is $2(1 - 2m)/9$, and no $\Delta/4$ loss occurs.

The coefficient convention matters. Put $\Theta = \langle \eta, \hat{\mathcal{T}} \rangle$. Conjugating entire witness and slot products when needed, the prime coefficients are fixed combinations

$$\nu(p)\mathbf{1}_{p \in 1_{\mathcal{T}}} = |\mathcal{T}|^{-1} \sum_{\vartheta \in \hat{\mathcal{T}}} (\nu\vartheta)(p), \quad \nu \in \Theta.$$

Profiles and zero extensions are conjugated with their factors. Large retained rows induce outside Θ , since they ramify at a good prime; bounded unit rows are handled separately.

For witness lengths $r < 1$, $m < 1/2$, the inverse and doubled plain capacities are $(1 - r)/2$ and $2(1 - 2m)/9$. The witnesses satisfy

$$r + m \geq t - O(\varepsilon), \quad r \leq t + O(1/\log U)$$

by [2, Proposition 8.3]. Selecting whole slots in decreasing amplitude gives the two count exponents

$$A_I(r) = 1 - \delta\{x + (1 - x)r\}, \quad S_t(r) = 1 - \delta\{4x/9 + (2 - 8x/9)(t - r)\}, \quad x = q/\delta.$$

The first decreases and the second increases with r . Their intersection gives

$$R_{\text{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x}(3/2 - t).$$

For $r \geq 1$, inverse amplification and the upper bound for r give $L(t) = 1 - \delta + (\alpha - \delta)(t - 1)$. The unweighted plain bound handles $m \geq 1/2$. Zero-capacity neighborhoods use these unweighted estimates, not a marked estimate with vanishing decrement. Thus the count is bounded by $\max\{R_{\text{short}}(t), L(t)\}$, with the requested small losses. The inequality $r \leq t + O(1/\log U)$ is essential here: it bounds the increasing long-inverse exponent, rather than merely asserting that one of the witnesses is large.

For $1/50 < \delta < \alpha$, balance these bounds as in [2, (20.8)] and the proof of [2, Lemma 20.2], by setting

$$\begin{aligned} y &= 1/2 - x, \quad D_x = (37 + 34y)/18, \quad P_x = (7 + 18y + 8y^2)/9, \\ \mathcal{J} &= (\alpha - \delta)D_x + \delta P_x = \{185 + 170y + (-138 + 12y + 96y^2)\delta\}/108, \\ t_* &= 1 + \frac{\delta P_x}{2\mathcal{J}}, \quad R_* = 1 - \delta + \frac{(\alpha - \delta)\delta P_x}{2\mathcal{J}}. \end{aligned} \tag{21}$$

Indeed $D_x(R_{\text{short}} - L) = \delta P_x/2 - \mathcal{J}(t - 1)$. Since $35/54 \leq \mathcal{J} \leq 5/2$, this yields uniformly

$$R \leq R_* + c_{\text{row}}, \tag{22}$$

with arbitrarily small prescribed witness, capacity and rounding loss c_{row} . At $\delta = \alpha$ use $R \leq 1 - \delta + c_{\text{row}}$; the floor $a = 51/100$ uses $R = 1$.

As in [2, Section 20.5], the new scales supply these moments: if $1/2 \leq d \leq h + \zeta$ and $0 < \zeta < 257/12000 = 5\ell - h$, then $\ell/d > 1/5 > 7/37$. Choose the moment mesh first, then sufficiently many even, disjoint physical slots. The plain moment's auxiliary prime pool remains separate. Their strict exponent gaps absorb annular $O(1/\log U)$ errors without changing any source mask.

5.2 An exact endpoint inequality

For sufficiently small $e > 0$, repeat the fixed-bin contour proof of [2, Lemma 20.1], applying [2, Lemmas 10.3 and 10.4] with $\sigma_0 = \theta$ (see Appendix A), on

$$(\operatorname{Re} s, \operatorname{Re} w, \operatorname{Re} z) = (a + 16e, 1 - a - 6e, 17/50).$$

The global line $\beta_* + 20e$ and buffered rectangles remain valid; $a \geq 51/100$ and $\operatorname{Re}(s + w) \geq 1 + 10e$ admit the unchanged first Euler region. Form amplitude and witness sets only on the retained contour. Joint conductor allocation bounds the numerator and all error slots together.

The actual Mellin and slot powers give, relative to $C(\theta)$, the first line of [2, (20.4)] with $7/8$ replaced by θ , that is, [2, (10.15)] with $\sigma_0 = \theta$ and $g = q\ell$:

$$E_\theta(d) = a - \theta + h(17/50 - 1/6) - al_y - \ell/2 + q\ell + d(R + \delta/2 - 17/50). \quad (23)$$

In particular $q\ell$ is already measured at base Z .

The next lemma re-runs the certificate of [2, Lemma 20.2] with the perturbation ϖ retained.

Lemma 5.1 (cf. [2, Lemma 20.2]). *For $0 \leq \delta \leq 5/6$, $0 \leq x \leq 1/2$, and $\varpi = 1/6000$, substitution of $q = \delta x$, $R = R_*$ gives $E_\theta(h) \leq -m_{\text{cert}}$, where $m_{\text{cert}} = 1/200000$.*

Proof. Write $E_0 = -1/48 + (2/3)\delta + \delta x/6 - (13/16)(1 - R_*)$; this is the quantity E_* of [2, Lemma 20.2], from the second line of [2, (20.4)]. The new geometry gives exactly

$$E_\theta(h) = E_0 + \varpi\{-1/4 + \delta(1 + x) + (3/2)R_*\}.$$

Multiplication gives $10368\mathcal{J}(-E_\theta(h) - m_{\text{cert}}) = A\delta^2 + B\delta + C$, where

$$\begin{aligned} A &= 6(32032y^3 + 94028y^2 + 131008y + 51021)/125, \\ B &= -(650644y^2 + 9423268y + 5918793)/3125, \\ C &= 6186(34y + 37)/625, \end{aligned}$$

and

$$4AC - B^2 = \frac{19787957489264y^4 + 69061294318816y^3 + 50726295758792y^2 + 10600196228952y + 1254989151}{9765625} > 0.$$

Since $A, \mathcal{J} > 0$, completing the square proves the claim (this discriminant test replaces the decomposition [2, (20.9)]): $4A(A\delta^2 + B\delta + C) = (2A\delta + B)^2 + 4AC - B^2 > 0$. $\square$

5.3 All moderate ranges

The remaining branches mirror [2, (20.5), (20.7) and (20.11)] at the perturbed geometry. The count envelopes have $1 - \delta \leq R \leq 3/2$; their frequency slope $R + \delta/2 - 17/50$ lies between $4/25$ and 2 . Thus $d = h$ controls $1/2 \leq d \leq h$; extending to $h + \zeta$ costs 2ζ . The other branches give

$$\begin{aligned} \text{floor:} & & E_\theta(h) &\leq -7/1200 + 15\varpi/8, \\ \delta = 5/6, R = 1 - \delta: & & E_\theta(h) &\leq -1/48 - \delta/16 + 15\varpi/8, \\ 1/100 \leq d \leq 1/2: & & E_\theta(d) &\leq -49/14400 + 139\varpi/150. \end{aligned}$$

The last uses $R = 76/75 - 2\delta/3$, whose slope is positive and whose geometry derivative is $a/2 + 1/100 + q \leq 139/150$. All three margins exceed m_{cert} .

If nominal row and frequency exponents d, v differ, with $v \leq h + \zeta$, $d - v \leq \mu$, their extra cost is at most $2\zeta + (3/2)\mu$. Including contour shifts, amplitude width ϑ , central power allowance ε_0 , and the remaining finite loss λ_{loss} , the total is bounded by

$$\Lambda = hc_{\text{row}} + 2\zeta + (3/2)\mu + 26e + (K + 8)\varepsilon_0 + \vartheta/5 + \lambda_{\text{loss}}. \quad (24)$$

Here 26 bounds $16 - 6l_y + 12v$, and $\ell < 1/5$. The compact parameter ranges and positive denominator bounds make every loss uniform. Choose their budgets before the target, the mesh before K , and ε_0 after K , so that $\Lambda < m_{\text{cert}}/4$. Reserving another $m_{\text{cert}}/4$ for the normalizer and finite dyadic multiplicities leaves a positive saving. Comparison with $C(\beta_*)$ subtracts the further positive gap Γ . This accounts for every moderate nonprincipal row.

6 Principal recovery and continuation

6.1 The principal correction below seven eighths

Set $a_0 = 1/24000$. The second Euler region D_2 of [2, Section 10.2 and (7.15)] is defined by $\text{Re } s \geq 7/8$, $\text{Re } w \geq 19/20$, $\text{Re } z \geq 33/200$. Since $\beta_* + e$ may now lie below $7/8$, we extend the local estimates of [2, Sections 7 and 16] to

$$\text{Re } s \geq \theta, \quad \text{Re } w \geq 19/20, \quad \text{Re } z \geq 33/200. \quad (25)$$

For the local ratios V, D, R of Section 7 of [2], the denominators remain separated from zero:

$$|V| \leq q_p^{-99/100}, \quad |D| \leq q_p^{-\theta}, \quad |R| \leq q_p^{-56/25+6a_0}.$$

For good primes the largest defect terms have exponents $4 - 6\text{Re } s - 6\text{Re } z$ and $1 - \text{Re } s - \text{Re } w - 6\text{Re } z$. For row primes the six-valuation table has least decay $1 - \text{Re } s - \text{Re } w$; the other potentially large exponents $3/2 - 3\text{Re } s$, $2 - 4\text{Re } s$ are smaller. Hence

$$|H_p - 1| \ll \begin{cases} q_p^{-363/200+a_0}, & p \nmid u, \\ q_p^{-33/40+a_0}, & p \mid u. \end{cases} \quad (26)$$

The constants are uniform in phases and heights. The first product converges normally, and the second costs q_u^ε by the divisor-product bound. Write $\mathcal{H}_{\eta,u} = \prod_{p \notin S} H_p$ for the unselected Euler product. At the principal residue $u = 1$ (so $j = 0$ and $\rho = 1$), $w = 1$, $z = 1/6$, the local identity [2, (7.11)–(7.12)] has $V = W = W_{\text{loc}} = q^{-1}$. Its completed part becomes $P_p^* = (1 + q^{-1})(R - D)/(1 - R)$, and $P_p = (1 + q^{-1})/(1 - q^{-1}) + P_p^*$. Hence [2, (7.12)] simplifies to

$$H_p(s) = \frac{1 - q^{-2}}{1 - R} \left(1 + \frac{q^{-1}(D - R)}{1 - D} \right), \quad D = \eta(p)q^{-s}, \quad R = \overline{\alpha(p)}^6 \eta(p)^6 q^{3-6s}, \quad q = q_p.$$

The same formula appears as [4, (8)], an AI-assisted adaptation of OpenAI's argument. For $\text{Re } s > 2/3$, the denominators are nonzero. So is the numerator, since $q \geq 7$ and

$$|(1 - q^{-1})D + q^{-1}R| < q^{-2/3} + q^{-2} < 1.$$

Moreover, $|H_p - 1| \ll q^{-2} + q^{-1-\text{Re } s} + q^{3-6\text{Re } s}$. The product therefore converges normally and gives

$$H_\eta(s) = \mathcal{H}_{\eta,1}(s, 1, 1/6) \quad \text{holomorphic and nonzero on } \text{Re } s > 2/3. \quad (27)$$

On each closed sub-half-plane it is bounded above and away from zero: use the local bounds on a finite head and the summable defects on the tail. This principal assertion needs no additional prime cutoff; the general contour estimates retain their fixed cutoff.

The exact cancellation identity [2, (16.8)] bounds $G_p + H_p$ by powers $-\theta, -99/100, -273/200 + 5a_0, -47/50$. At the principal residue, where H_p is bounded away from zero, this gives

$$B_p = G_p/H_p = -1 + O(q_p^{-4/5}). \quad (28)$$

General contours still use the quotient-free tuple correction.

6.2 Recovering the same signal

This subsection follows [2, Section 20.1] at the new scales. Put

$$S_i(Z) = \sum_{p \in \mathcal{P}_i(Z)} W_i(q_p/P_i) q_p^{-5/6}, \quad A_{\mathcal{T}}(Z) = (-1)^K Z^{-\ell/6} \prod_i S_i(Z).$$

The fixed-ray asymptotic of [2, Lemma 13.1] gives

$$S_i(Z) \sim \frac{P_i^{1/6}}{|\mathcal{T}| \log P_i} \int_0^\infty W_i(y) y^{-5/6} dy > 0.$$

Thus, for even K , $A_{\mathcal{T}}(Z) > 0$ eventually and $A_{\mathcal{T}}(Z) \asymp (\log Z)^{-K}$. Fixed exclusions affect only the threshold; its inverse costs one small power. With the unchanged positive residue constant

$$c_S = \widehat{W}_1(1) M(1/6) (\text{Res}_{v=1} \zeta_F^S(v))^2 / 6,$$

define

$$\begin{aligned} J_\eta(Z) &= \mathcal{I}_{\eta, \text{mod}}(Z) / \{c_S A_{\mathcal{T}}(Z)\}, \\ f_\eta(Z) &= \frac{1}{2\pi i} \int_{(2)} Z^{s-11/16} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds. \end{aligned} \quad (29)$$

Neither function contains a height cutoff. By (18), for every $\varepsilon_{\text{low}} > 0$,

$$|J_\eta(Z)| \ll_{\eta, \varepsilon_{\text{low}}} Z^{C(\theta) + \varepsilon_{\text{low}}}. \quad (30)$$

For $u = 1$, keep $\text{Re } s = \beta_* + e$, cross $w = 1$ to $19/20$, then $z = 1/6$ to $33/200$. Repeating the residue proof of [2, Lemma 10.5] in (25) crosses only the two scalar zeta poles. Its geometric savings, the analogues of [2, (20.3)], are

$$m_w = l_y/20 = 5749/240000, \quad m_z = h/600 = 3251/2400000.$$

The latter includes all selected-slot powers. At the double residue, $X^{1/3} Z^{s-5/6} = Z^{-\ell/6} Z^{s-11/16}$; (28) gives relative error $O(Z^{-\kappa_P})$, with $\kappa_P = (2/5) \min_i \ell_i > 0$. Move the main integral right to 2, bounding the slot error on its original line. The Gaussian pays the horizontal joins. The principal recovery error is

$$\ll Z^{C(\beta_*) + (1+h)e - m_w + \varepsilon_{\text{pr}}} + Z^{C(\beta_*) + e - m_z + \varepsilon_{\text{pr}}} + Z^{C(\beta_*) + e - \kappa_P + \varepsilon_{\text{pr}}}. \quad (31)$$

6.3 Outer rows and height closure

This subsection repeats [2, Lemma 10.6 and Section 20.3]. For $d < 1/100$, use $\operatorname{Re} s = 7/8 + 8e$, to the right of β_* by [2, Theorem 1.1], and $(\operatorname{Re} w, \operatorname{Re} z) = (1/2, 17/50)$. OpenAI uses the line $\operatorname{Re} s = \beta_* + e$, which lies in the first Euler region $D_1(3/8)$ of [2, Section 10.2] only when $\beta_* + e \geq 7/8$. Our line has $\operatorname{Re}(s + w) = 11/8 + 8e$, so it lies in $D_1(3/8)$. The proof of [2, Lemma 16.2] on its first set of lines uses only this region and $\operatorname{Re} s \geq 7/8 + e$, so its local bounds apply unchanged; the shift to the right of $\beta_* + e$ costs $7/8 - \beta_* + 7e$. The full absolute tuple bound, including selected primes dividing u , is $q_u^\varepsilon Z^{17\ell/50}$; the row exponent is $63/50 < 2$. Paying $7/8 - \beta_* < \varpi/4$ leaves saving

$$m_{\text{small}} = 63/800 - 51\varpi/100 - \varpi/4 > 1/16.$$

This includes bounded nontrivial units and principal denominators. In the absolute estimate, $G_p = O(1)$ off u and $G_p = O(q_p^{1/2})$ on u . The latter choices are divisor-bounded, and their actual weight $q_p^{\operatorname{Re} z - 1}$ absorbs this growth [2, Lemma 16.2]. These ramified selections therefore remain in the tuple sum. For the full large-row contributions $\mathcal{R}_\eta(U; Z)$ in (3), absolute lines $(2, 2, z_\infty)$ give

$$\sum_{U > Z^{h+\zeta}} |\mathcal{R}_\eta(U; Z)| \ll Z^{B_0 + (h+\zeta)(1+\varepsilon) - \zeta z_\infty + \varepsilon}, \quad B_0 = l_x/2 + 1 + l_y.$$

Choose z_∞ after ζ to obtain the required saving.

Choose $\varepsilon_{\text{low}} = \Gamma/2$. Before fixing a target, choose moment losses, the mesh and disjoint slots, then the amplitude and contour widths so that (24) holds and

$$(1 + h)e + \varepsilon_{\text{pr}} < m_w/4, \quad e + \varepsilon_{\text{pr}} < \min\{m_z, \kappa_P\}/4.$$

Reserve fractions of the remaining savings for normalizing and summing dyads. Choose ζ, z_∞ and the prime cutoff next. For each target fix its arithmetic data and internal seminorm and Sobolev orders. The common height allocation of [2, Sections 10 and 20.6] then gives

$$|J_\eta - f_\eta| \ll_{\eta, N} Z^{C(\beta_*) - m_1} (1 + T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N}, \quad T_1 = Z^{\tau_\eta},$$

where $m_1 > 0$ is common to all targets, and A_η, B_η are finite and independent of the later external order N . Choose $\tau_\eta > 0$ to meet the detector ceiling and $\tau_\eta A_\eta < m_1/4$, then N_η large enough to pay all discarded axes and joins, and finally the threshold. Increasing external orders changes only constants and thresholds. Thus some common $\sigma > 0$ satisfies

$$|J_\eta| \ll_\eta Z^{C(\theta) + \varepsilon_{\text{low}}}, \quad |J_\eta - f_\eta| \ll_\eta Z^{C(\beta_*) - \sigma}. \quad (32)$$

6.4 Continuation

The continuation repeats the proof of [2, Proposition 2.1] with $\sigma_0 = \theta$ and $\omega = \varepsilon_{\text{low}}$. Its hypothesis [2, (2.2)], $\sup |H_\eta - 1| \leq 1/2$, is replaced by (27) and the two-sided bounds after it, which are all that proof uses of H_η . The passage to the full family is [2, Proposition 11.3] with $\sigma_0 = \theta$; we repeat its short proof.

Set $\epsilon_* = \min\{\Gamma - \varepsilon_{\text{low}}, \sigma\} > 0$. Then $f_\eta(Z) = O_\eta(Z^{C(\beta_*) - \epsilon_*})$ at infinity. At zero it decays faster than every fixed power, by shifting its integral to any $B > 2$, using the bounded reciprocal Euler product, (27), and the Gaussian. Consequently

$$F_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-C(s)} \frac{dZ}{Z}$$

is holomorphic for $\operatorname{Re} s > \beta_* - \epsilon_* > \theta$. Fourier inversion with $Z = e^u$ identifies it on $\operatorname{Re} s > 1$ as $e^{(s-5/6)^2} H_\eta(s) / L_F^S(s, \eta)$. Since H_η is holomorphic and nonzero, this continues the reciprocal holomorphically to the common half-plane $\operatorname{Re} s > \beta_* - \epsilon_*$. Deleted Euler factors are nonzero there, while the definition of β_* supplies a primitive target with a zero in that half-plane. This contradiction proves $\beta_* \leq \theta$. The saving ϵ_* is common to the family; it is precisely this uniformity, rather than a common lower threshold or implied constant, that lets us use the zero supremum. The argument places no bound on the imaginary part of the excluded zero.

This proves the Hecke assertion for primitive characters. An induced presentation differs by finitely many factors $1 - \eta_0(\mathfrak{p})q_{\mathfrak{p}}^{-s}$, which are nonzero for $\operatorname{Re} s > 0$. The same boundary therefore holds for all finite-order Hecke presentations.

For a Dirichlet character χ , form the finite-order Hecke character $\eta_\chi(\mathfrak{a}) = \chi(N\mathfrak{a})$. Delete a finite rational prime set $S_{\mathbb{Q}}$ containing 3 and the modulus support, and let S_F consist of all primes above it. The split and inert Euler factors give

$$L_F^{S_F}(s, \eta_\chi) = L^{S_{\mathbb{Q}}}(s, \chi) L^{S_{\mathbb{Q}}}(s, \chi\chi_{-3}).$$

On $\theta < \operatorname{Re} s < 1$, both factors on the right are holomorphic and their product is nonzero. Restoring the finite Euler factors proves Dirichlet nonvanishing there. For $\operatorname{Re} s \geq 1$, use the classical nonvanishing of Dirichlet L -functions on $\operatorname{Re} s \geq 1$ [3, Chapters 13–14], with principal poles excepted. This also covers imprimitive Dirichlet characters and, in particular, the zeta function.

Finally, the functional equations reflect a critical-strip zero of a primitive character to a zero of its conjugate character with real part $1 - \operatorname{Re} \rho$. The family is closed under conjugation. The deleted factors have no zeros in the critical strip, so every member has the lower boundary $1 - \theta = 3001/24000$. This proves Proposition 1.2.

7 Two finite adaptive steps

The first-stage boundary can itself be used in the plain moment. We now make two such uses, each with a strict endpoint margin. The arithmetic construction and the recovery source remain those of [2]; the extension of the plain moment is proved in [1, Theorem 1.1].

7.1 The extended moment and its physical consumer

Lemma 7.1 (Extended plain moment). *Retain all hypotheses and notation of [2, Lemma 18.1]. For $13/18 \leq \kappa \leq 3/4$, assume the global family bound $\beta_* \leq (1 + \kappa)/2$. Let the element rows satisfy $0 < q_k \ll Z^m$, with $\psi_k(n) = \tau(n)\chi_n(k)$, and put $M = m + q$, where q bounds the displayed moving conductor union before cancellation. For positive total prime length z , assume $n_1 + n_2 + 6\kappa z \leq M$ and restrict to inducing characters outside the prescribed finite group Θ . Then*

$$\sum_{k \in \mathcal{R}_z} \left| S_{\psi_k}(n_1; W_1) S_{\psi_k}(n_2; W_2) \prod_i Q_{\psi_k, i} \right|^2 \ll Z^{M+\varepsilon}.$$

At $z = 0$, bounded plain lengths are unrestricted and the rows remain nonprincipal. The slot mesh precedes the fixed slot count, uniformly on bounded length ranges. Both plains and every slot use the same puncture, fixed within the row sum. All defining-modulus zeros, including canceled and six-divisible factors, remain. Prime supports are disjoint before masking, and their coefficients are fixed finite combinations from Θ , independent of the varying row. The original label ranges, finite seminorms and polynomial separated-height costs are retained.

The proof in [1] follows every recursive child of [2, Section 18]. The global input licenses the prime contour $(1 + \kappa)/2 + e_{\text{an}}$ for each nonprincipal child; it is not a bound inferred from its parent's detector bin. The changed terminal length is $z \leq 3M/13$. The comparison has admission reserve $M/15$; its clipping errors are $8e_{\text{cl}}/5$ and $13e_{\text{cl}}/5$, so choosing $\xi + 13e_{\text{cl}}/5 < M/15$ preserves that reserve. Natural zeros, the common Fourier measure, exceptional equal-product cancellations and the backward seminorm allocation are unchanged.

For the physical rows the twist is fixed-ray, so there is no additional moving radical in M . Write $\delta = 2a - 1$, $q = \delta x$, $0 \leq x \leq 1/2$, and $\alpha = 5/6$. The actual detector witnesses obey

$$r + m \geq t - O(\varepsilon), \quad r \leq t + O(1/\log U), \quad 1 \leq t \leq 3/2.$$

For $r < 1, m < 1/2$, the inverse and doubled-plain capacities are $(1 - r)/2$ and $(1 - 2m)/(6\kappa)$. Put

$$c = \frac{1}{3\kappa}, \quad B = 2 - 2cx, \quad D = B + 1 - x, \quad P = B(1 - x), \quad \mathcal{J} = (\alpha - \delta)D + \delta P.$$

Their count exponents are $1 - \delta\{x + (1 - x)r\}$ and $1 - \delta\{cx + B(t - r)\}$. Balancing them and the long-inverse branch gives

$$\begin{aligned} R_{\text{short}}(t) &= 1 - \delta + \delta P(3/2 - t)/D, & L(t) &= 1 - \delta + (\alpha - \delta)(t - 1), \\ t_* &= 1 + \frac{\delta P}{2\mathcal{J}}, & R_* &= 1 - \delta + \frac{(\alpha - \delta)\delta P}{2\mathcal{J}}. \end{aligned} \tag{33}$$

These formulas use both witness inequalities, not an equality for their lengths. Uniformly $D \geq 2$, $\mathcal{J} \geq 25/39$, and the inverse crossing has $r_* \geq 23/37$. Thus the existing inverse prime supply $7/37$ suffices; the plain supply is at most $c/6 \leq 1/13$. Zero-capacity neighborhoods use the unweighted estimates. Whole-product conjugation preserves the physical coefficients and zeros. Rowwise selected real parts and heights are recovered by the same common parameter-Sobolev measure, not by row-dependent prime coefficients [1, Proposition 4.1].

7.2 Two exact endpoint certificates

Set $\theta_0 = 20999/24000$, supplied by Proposition 1.2. For each of the following stages use $\kappa = 2\theta_{i-1} - 1$, and the geometry

$$\ell_i = 1/6 + p_i, \quad h_i = 13/16 + 3p_i/2, \quad l_{x,i} = 17/48 - p_i/2, \quad l_{y,i} = 23/48 - p_i/2.$$

Here ℓ_i is the total prime length for stage i , not an individual slot length. The latter are fixed after the moment mesh.

Stage	κ	c	p_i	m_i	$\theta_i = 7/8 - p_i/4$
1	8999/12000	4000/8999	1/5836	10^{-8}	20425/23344
2	8753/11672	11672/26259	107121/625000000	10^{-9}	2187392879/2500000000

Both κ 's lie in $[13/18, 3/4]$. At stage 2 the prime contour is $\theta_1 + e_{\text{an}}$, licensed by the already established stage 1, not by the output θ_2 .

The principal exponent remains $C(s) = s - 11/16$. The literal central endpoint, with the count (33), is

$$E_i = -1/48 + (2/3)\delta + \delta x/6 - (13/16)(1 - R_*) + p_i\{-1/4 + \delta(1 + x) + (3/2)R_*\}.$$

We give a compact rational certificate on the full rectangle $0 \leq \delta \leq 5/6$, $0 \leq x \leq 1/2$. Set $y = 1/2 - x$ and

$$\begin{aligned} D &= 5/2 - c + (1 + 2c)y, & P &= 1 - c/2 + 2y + 2cy^2, \\ k_i &= 1/48 - 5p_i/4 - m_i, & v_i &= 1/16 + y/6 + p_i y. \end{aligned}$$

Then exact multiplication gives

$$\begin{aligned} 2\mathcal{J}(-E_i - m_i) &= A_i\delta^2 + B_i\delta + C_i, \\ A_i &= 2(P - D)v_i + h_i P, \\ B_i &= 2\alpha D v_i + 2(P - D)k_i - h_i \alpha P, & C_i &= 2\alpha D k_i. \end{aligned} \tag{34}$$

For both rows of the table, rational expansion of these displayed formulas gives $[y^j]A_i \geq 1/4$ for $0 \leq j \leq 3$, and

$$[y^j](4A_i C_i - B_i^2) > d_j \quad (0 \leq j \leq 4), \quad (d_0, \dots, d_4) = (10^{-8}, 1/25, 19/100, 13/50, 3/40).$$

All bounds are rational coefficient inequalities. In particular $A_i > 0$ and $4A_i C_i - B_i^2 > 0$ for $y \geq 0$. The identity $4A_i(A_i\delta^2 + B_i\delta + C_i) = (2A_i\delta + B_i)^2 + 4A_i C_i - B_i^2$ proves $E_i < -m_i$. Stage 2 therefore retains the full 10^{-9} margin; no sampling or decimal comparison is used.

7.3 Activation at the new scales

For each stage repeat the same compensated probe (1), including every marked subset, the original disjoint windows and the full quotient-free correction. The rescaled minima are $M' \geq 1/2 - 3p_i > 0$, $l_{x,i} - d \geq 3/16 - 3p_i/2 > 0$, and $l_{y,i} - d - 11b/6 \geq 1/12 - 3p_i/2 > 0$. Lemma 3.1 and the physical Gram recovery apply since $\ell_i < 1/5$; the tuple payment remains

$$-d + (5\ell_i - 1 + d)_+/8 \leq -7d/8.$$

Thus the complete low bound is $Z^{3/16 - p_i/4 + \varepsilon} = Z^{C(\theta_i) + \varepsilon}$, with the same finite profile, cusp, mask and height costs.

For moderate rows use the unchanged $D_1(10e)$ contours and (33). The floor, terminal and intermediate branches retain the bounds

$$-7/1200 + 15p_i/8, \quad -1/48 - \delta/16 + 15p_i/8, \quad -49/14400 + 139p_i/150,$$

respectively; each has a margin larger than m_i . The frequency slope is between $4/25$ and 2 . Choose $0 < \zeta < 5\ell_i - h_i = 1/48 + 7p_i/2$; then $\ell_i/d > 1/5 > 7/37$ for $1/2 \leq d \leq h_i + \zeta$. Thus the fixed disjoint physical slots provide the required capacities. Allocate the aggregate losses in (24) below $m_i/8$: choose the mesh before the even fixed slot count, then the allowances involving that count. The strict denominator and selection margins make this allocation uniform in the targets.

The second Euler box is now $(\operatorname{Re} s, \operatorname{Re} w, \operatorname{Re} z) \geq (\theta_i, 19/20, 33/200)$. With $a_0 = p_i/4$, the good- and row-prime defects in (26) are $q_p^{-363/200 + a_0}$ and $q_p^{-33/40 + a_0}$; the denominator decay is $56/25 - 3p_i/2 > 2$. These give normal convergence and divisor-product control as before. The selected-factor error exponents $-\theta_i, -99/100, -273/200 + 5p_i/4, -47/50$ are all at most $-4/5$. Hence the same principal correction, nonvanishing H_η , and logarithmic normalizer are retained. Its residue reserves are $l_{y,i}/20, h_i/600, (2/5) \min_j \ell_{i,j} > 0$. Normalization losses are reserved after the slots are fixed.

Small rows use $(\operatorname{Re} s, \operatorname{Re} w, \operatorname{Re} z) = (7/8 + 8e, 1/2, 17/50)$, which lies in $D_1(3/8)$ and is right of every target zero by the input theorem. Under $\beta_* > \theta_i$, its shift from $\beta_* + e$ costs less than

$p_i/4 + 7e$. Together with the scale cost $51p_i/100$, the saving is $63/800 - 19p_i/25 > 1/16$, including bad-place and nontrivial-unit rows. No use of $\beta_* + e$ in $D_1(3/8)$ below $7/8$ is made. For large rows, $B_{0,i} = 53/32 - 3p_i/4$. After fixing $\zeta > 0$, choose a fixed absolute contour order v so that $B_{0,i} + (h_i + \zeta)(1 + \varepsilon) - \zeta v + \varepsilon$ is below the required exponent. These estimates retain all axes, joins and rescaled profiles. For each target the internal finite orders precede its height exponent τ_η , which precedes the later external tail order. Constants and thresholds may depend on the target; the remaining power reserve is common.

7.4 Finite feedback and the full family

We finish the proof of Theorem 1.1. At stage i , assume $\theta_i < \beta_* \leq \theta_{i-1}$, put $\Gamma_i = \beta_* - \theta_i$, and choose the low allowance $\omega = \Gamma_i/2$. All slots and power reserves precede the individual target. The recovery of Section 6, with the foregoing parameters, produces the same signal (29) and a common $\sigma_i > 0$ such that

$$|J_\eta(Z)| \ll_\eta Z^{C(\theta_i)+\omega}, \quad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\sigma_i}.$$

The uniform logarithmic control used in this recovery comes from the input's all-height zero location and retains its conductor and height logarithms. It does not require uniform target-dependent thresholds. Let $\epsilon_i = \min(\Gamma_i/2, \sigma_i) > 0$. The Mellin transform of f_η continues $e^{(s-5/6)^2} H_\eta(s)/L_F^S(s, \eta)$ holomorphically to $\operatorname{Re} s > \beta_* - \epsilon_i$, exactly as in the previous section. The numerator is nonzero there. A primitive target zero sufficiently near the supremum contradicts this continuation; attainment of the supremum is unnecessary. This proves $\beta_* \leq \theta_i$. Starting with Proposition 1.2 proves stage 1 and then stage 2, without a contour at an unproved output boundary.

Finite induced Euler factors are nonzero for $\operatorname{Re} s > 0$, so the same result holds for every finite-order Hecke presentation. For a Dirichlet character χ , its norm lift $\eta_\chi(\mathfrak{a}) = \chi(N\mathfrak{a})$ satisfies, after deleting the modulus support and 3,

$$L_F^{S_F}(s, \eta_\chi) = L^{S_Q}(s, \chi) L^{S_Q}(s, \chi\chi_{-3}).$$

Both factors are holomorphic on $\theta_2 < \operatorname{Re} s < 1$; their nonzero product and finite restoration give the Dirichlet assertion there. The coarse input covers $\operatorname{Re} s \geq 1$, with principal poles excepted. There is no height restriction. Functional equations and closure under conjugation reflect critical-strip zeros, yielding the lower boundary $1 - \theta_2 = 312607121/2500000000$. This proves the full theorem. The two strict gains are $41/35016000$ and $39539/3647500000000$; only these two finite steps are used.

8 The limit of the parameter refinement at $\kappa = 3/4$

With the asymmetry fixed at $b = 1/8$ and the plain-moment exponent fixed at $\hat{\kappa} = 3/4$, the perturbation is close to the limit of this particular balanced witness estimate. At $y = 0$ the zero-margin quadratic is

$$Q_\varpi(\delta) = (2448 + 6048\varpi)\delta^2 + (-1896 + 11520\varpi)\delta + 370 - 22200\varpi.$$

Its discriminant test gives

$$4AC - B^2 = -576(1162800\varpi^2 + 286020\varpi - 49).$$

The vertex remains inside $0 \leq \delta \leq 5/6$ near the present perturbation. Therefore, at $\hat{\kappa} = 3/4$, this estimate cannot support a larger perturbation past

$$\varpi_* = -\frac{1589}{12920} + \frac{7\sqrt{465053}}{38760} = 0.0001711975385\dots,$$

whose loss-free crossing $7/8 - \varpi_*/4$ is approximately 0.8749572006. This is the limit of the retained endpoint inequality at $\hat{\kappa} = 3/4$ only. It is not a lower bound for the true physical source, for the same method with other plain-moment exponents, or for other arithmetic methods.

The adaptive argument in Section 7 passes this fixed- (b, κ) ceiling by using the previously established boundary in the prime-slot capacity. Of its gain of about $1.18 \cdot 10^{-6}$ over θ_0 , about $1.13 \cdot 10^{-6}$ is already available at the parameter limit above. The finite feedback approaches the loss-free value 0.87495715035054; that limit is not asserted in Theorem 1.1. Further decimal tuning within this estimate would give little additional improvement. A substantial advance needs a stronger arithmetic estimate.

Supplementary verification

The supplementary archive (repository commit to be pinned on release) builds on OpenAI’s Lean formalization of [2], at the commit pinned in the bibliography; its definitions and mathematical input modules are unchanged. Its final theorem is the standard Mathlib statement `riemannZeta s ≠ 0` for $20999/24000 < \text{Re } s$. This statement passed exact comparison against the target and replay in the Lean kernel, with axioms exactly `propext`, `Classical.choice` and `Quot.sound`. The corresponding Hecke and Dirichlet declarations compiled but were not comparator-checked. The archive also contains exact rational checks of the first-stage endpoint certificate. It verifies θ_0 , not the adaptive boundary θ_* . The latter has the conventional proof above and the recursive proof in [1], without a new Lean, Comparator or kernel replay. The archive accompanies and does not replace these proofs.

Attribution and preparation

The cubic-theta construction, the unchanged arithmetic inputs and the quadratic endpoint-certificate technique [2, Lemma 20.2] are due to OpenAI [2], as is the Lean formalization on which the supplementary archive builds. The contribution here is the modified-scale low estimate, which extends [2, Lemma 15.1 and Proposition 15.3] to the perturbed geometry; the re-run of the endpoint certificate with the perturbation retained; the extension of the principal Euler region below $7/8$; and the recovery at the new boundary; the extended plain-moment range and its finite adaptive reuse, proved in [1] and Section 7. AI tools assisted with the derivations, algebraic checks and drafting.

Contemporaneous refinements. Liu [5] and ProofCouncil [6] report smaller boundaries. Proof-Council also extends the plain moment to $\kappa \geq 13/18$ and jointly optimizes the asymmetry b , which is fixed at $1/8$ here. The adaptive extension and witness balancing therefore overlap; their gains cannot be added to those of this paper. This paper records a conditional derivation and its exact certificates. It claims neither the smallest known boundary nor priority for the shared extension.

A Hypothesis audit

For each statement of [2] used here, the table records its printed hypotheses, the values used in this paper, and why its proof transfers. Part II of [2] works under the supposition [2, (12.1)], $\Delta = \beta_* - 7/8 \in (0, 1/24]$ and $\kappa = 2\beta_* - 1$, with the geometry [2, (12.4)], $\ell = 1/6$. For the first stage, by contrast, $\theta_0 < \beta_* \leq 7/8$, with $\sigma_0 = \theta$, $\hat{\kappa} = 3/4$ and the geometry (2), $\ell = 1/6 + \varpi$. The second Euler region D_2 of [2, Section 10.2] requires $\text{Re } s \geq 7/8$; it is replaced by the box (25). The table concerns that first stage. The adaptive stage additionally uses Lemma 7.1 and its complete proof in [1]; Section 7 records the changed capacity, endpoint and recovery payments.

Statement	Printed hypotheses	Used here	Why the proof transfers
Thm. 1.1	None (unconditional)	$\beta_* \leq 7/8$, as a hypothesis	Used as stated; this is the main reason the theorem is conditional.
Prop. 2.1	$\sigma_0 \in (1/2, 1)$, $\Delta_0 = \beta_* - \sigma_0 > 0$; common $0 < \omega < \Delta_0$, $\sigma > 0$; $\sup_{\text{Re } s > \sigma_0} H_\eta - 1 \leq 1/2$ (2.2); (2.4), (2.5)	$\sigma_0 = \theta$, $\omega = \varepsilon_{\text{low}} = \Gamma/2$; H_η of (27)	The proof uses (2.2) only for boundedness and nonvanishing of H_η ; Section 6 supplies both on every closed sub-half-plane of $\text{Re } s > 2/3$.
Prop. 11.3	$\sigma_0 \in [1/2, 1)$; primitive Hecke L zero-free on $\text{Re } s > \sigma_0$	$\sigma_0 = \theta$	Applied as stated.
Lemmas 4.5, 4.6, 4.10	Smooth profiles; fixed $\sigma_0 > 0$ in Lemma 4.10	As stated	No zero-free or geometry hypothesis.
Lemma 4.9	$a \in [1/2, 1]$, $0 < e < 10^{-3}$, zero-free disk; global form for $b \geq \beta_*$, $v > 0$	$a \geq 51/100$; $b = \beta_*$ or $b = 7/8 + 8e$	Applied as stated; the global form needs only $b \geq \beta_*$.
Lemma 7.1, (7.11)–(7.12); (16.3), (16.8)	Regions (7.14) and (7.15), the latter with $\text{Re } s \geq 7/8$	Box (25), $\text{Re } s \geq \theta$	The local defects remain summable there by (26) and (28).
Lemmas 10.3, 10.4	Def. 10.1 with $D_1(\epsilon_0)$, D_2 ; $\sigma_0 \in [7/8, 1)$, $\beta_* > \sigma_0$; $z_0 = 17/50$	$\sigma_0 = \theta$, $\theta < \beta_* \leq 7/8$; geometry (2)	Central contours lie in $D_1(10e)$, which does not involve $7/8$; σ_0 enters only as the reference $C(\sigma_0)$ in (10.15); the bin ceiling $a \leq \beta_*$ is unchanged.
Lemma 10.5	$\sigma_0 \in [7/8, 1)$, $\beta_* > \sigma_0$; (10.17) on a box in D_2 ; (10.18)	$\sigma_0 = \theta$; the same box, now inside (25)	D_2 enters only through holomorphy and (10.17), proved on the enlarged box; (10.18) holds with $\mu = \kappa_P$ by (28).
Lemma 10.6	$\sigma_0 \in [7/8, 1)$, $\beta_* > \sigma_0$; (10.20) on $(\beta_* + e, 1/2, 17/50)$	Small rows on $(7/8 + 8e, 1/2, 17/50)$; large rows on $(2, 2, z_\infty)$	The line is right of β_* by Thm. 1.1 and lies in $D_1(3/8)$; its extra cost $7/8 - \beta_* + 7e$ is paid from m_{small} .
Lemma 11.1	$\sigma_0 \in (1/2, 1)$, $\Delta_0 > 0$; common $m > 0$, $\omega < \Delta_0$	$\sigma_0 = \theta$	Applied as stated.
Lemma 14.3; Prop. 15.2	Bounded log-lengths; $Q, Y' \geq 1$, $P_a = Y'^2/Q \geq 1$	Perturbed scales	Hypotheses verified by (16).
Lemma 15.1	Geometry (12.4), $\ell = 1/6$	$0 \leq \ell \leq 1/5$	Proof repeated in Lemma 3.1; ℓ enters only through $N_* = 1 + \ell'$ and $z_a \leq \ell'$.
Prop. 15.3	Geometry (12.4), $\ell = 1/6$; exponent $3/16$	Geometry (2)	Proof repeated in Proposition 4.1; the extra term $(5\ell - 1 + d)_+/8$ is paid by the tuple count, since $\ell < 1/5$.
Lemma 16.2	First lines $(\beta_* + e, 1/2, 17/50)$ with $\beta_* > 7/8$	$(7/8 + 8e, 1/2, 17/50)$	The proof uses only $\text{Re } s \geq 7/8 + e$ and $D_1(3/8)$.
Prop. 8.3; Lemmas 17.1, 17.6	Detector bins; bounded ranges; strict moment inequalities	As stated, with slots from $\ell = 1/6 + \varpi$	No zero-free hypothesis; slot lengths enter only through bounded ranges and the mesh.
Lemma 18.1	$3/4 \leq \kappa \leq 1$; if $z > 0$ and $\kappa < 1$, then $\beta_* \leq (1 + \kappa)/2$	$\kappa = 3/4$; $\beta_* \leq 7/8$ by Thm. 1.1	Applied as stated.
Prop. 19.2	Under (12.1): $\Delta > 0$, $\kappa = 2\beta_* - 1$; $\alpha = 5/6$; prime supply above $7/37$; loss $+\Delta/4$	$\hat{\kappa} = 3/4$; supply $\ell/d > 1/5$	The proof uses κ only through $\delta \leq \kappa$ and Lemma 18.1, both supplied by Thm. 1.1; the loss came from $6\kappa = 9/2 + 12\Delta$ and vanishes.

Statement	Printed hypotheses	Used here	Why the proof transfers
Lemma 20.1	Lemma 10.4 with $\sigma_0 = 7/8$; (12.2); geometry (12.4)	$\sigma_0 = \theta$; exponent (23)	Only the affine exponent (20.4) changes; the contour and conductor-allocation steps use the slot system, not ℓ .
Lemma 20.2	Geometry (12.4), $\varpi = 0$; $0 \leq \delta \leq 5/6$, $0 \leq x \leq 1/2$	$\varpi = 1/6000$	Same algebra with ϖ retained; Lemma 5.1 gives margin $1/200000$.

Adaptive-stage hypotheses. At the first adaptive step, the input boundary is θ_0 and $\kappa_0 = 2\theta_0 - 1$; at the second it is the established θ_1 and $\kappa_1 = 2\theta_1 - 1$. Both lie in $[13/18, 3/4]$. The global bound $\beta_* \leq (1 + \kappa_j)/2$ comes from the previous stage, including its all-height and full-family continuation. The common masks, character zeros, finite slot coefficients and moving-radical width remain those of the extended moment. The small-row contour stays at $7/8 + 8e$, and its additional shift cost is paid explicitly. Thus no use of the output boundary is hidden in a prime estimate, recursive child or contour choice.

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